

Question 1 :

1. c

4. c

7. a/b

10. a

2. a

5. b

8. a

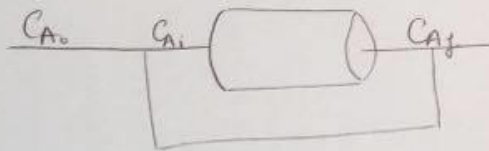
3. a

6. b

9. b

Question 2

1) Mole balance



$$\frac{dF_A}{dV} = r_A = -kC_A^2$$

$$F_A = 20C_A \Rightarrow 20 \frac{dC_A}{dV} = \frac{dF_A}{dV}$$

$$\Rightarrow 20 \frac{dC_A}{dV} = -kC_A^2$$

$$\Rightarrow \frac{dC_A}{C_A^2} = \frac{-k}{20} dV$$

$$\Rightarrow \int_{C_{A_i}}^{C_{A_f}} \frac{dC_A}{C_A^2} = \int_0^V \frac{-k}{20} dV$$

$$\Rightarrow -\frac{1}{C_A} \Big|_{C_{A_i}}^{C_{A_f}} = \frac{-k}{20} V$$

$$\Rightarrow -\frac{1}{C_{A_f}} + \frac{1}{C_{A_i}} = \frac{-kV}{20}$$

$$C_{A1} = \frac{C_{A0} + C_{A1}}{2}$$

$$X = 3/4$$

$$C_{A1} = 1/4 C_{A0}$$

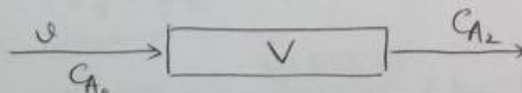
$$C_{A1} = \frac{C_{A0} + 1/4 C_{A0}}{2} = \frac{5}{8} C_{A0}$$

$$\Rightarrow -\frac{1}{\frac{1}{4} C_{A0}} + \frac{1}{\frac{5}{8} C_{A0}} = \frac{-kV}{2v}$$

$$\Rightarrow \frac{12}{5 C_{A0}} = \frac{kV}{2v}$$

$$\Rightarrow \frac{kV}{v} C_{A0} = \frac{24}{5}$$

b) No recycle



$$v \frac{dC_A}{dV} = \frac{dF_A}{dV} = -kC_A^2$$

$$\Rightarrow -\frac{1}{C_{A2}} + \frac{1}{C_{A0}} = \frac{-kV}{v}$$

$$\Rightarrow -\frac{1}{C_{A2}} = -\frac{1}{C_{A0}} - \frac{kV}{v} = -\frac{1}{C_{A0}} - \frac{24}{5 C_{A0}} = -\frac{29}{5 C_{A0}}$$

$$\Rightarrow C_{A2} = \frac{5}{29} C_{A0}$$

$$\Rightarrow X_A = \frac{24}{29}$$

Question 3 :

1)

$$\bullet \frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_3[ES][S] + k_{-3}[SES] - k_2[ES] + \beta k_2[SES] = 0$$

$$\Rightarrow k_1[E][S] + k_{-3}[SES] + \beta k_2[SES] = (k_{-1} + k_3[S] + k_2)[ES]$$

$$\Rightarrow [ES] = \frac{k_1[E][S] + k_{-3}[SES] + \beta k_2[SES]}{k_{-1} + k_3[S] + k_2} \quad (1)$$

$$\bullet \frac{d[SES]}{dt} = k_3[ES][S] - k_{-3}[SES] - \beta k_2[SES] = 0$$

$$\Rightarrow [SES] = \frac{k_3[ES][S]}{k_{-3} + \beta k_2} \quad (2)$$

$$\bullet (1) + (2) \Rightarrow [ES] = \frac{k_1[E][S] + \frac{k_3[ES][S]}{k_{-3} + \beta k_2} (k_{-3} + \beta k_2)}{k_{-1} + k_3[S] + k_2}$$

$$\Rightarrow [ES] = \frac{k_1[E][S] + k_3[ES][S]}{k_{-1} + k_3[S] + k_2}$$

$$\Rightarrow [ES] (k_{-1} + k_3[S] + k_2) - k_3[ES][S] = k_1[E][S]$$

$$\Rightarrow [ES] = \frac{k_1[S][E]}{k_{-1} + \cancel{k_3[S]} + k_2 - \cancel{k_3[S]}} \quad (3)$$

$$\bullet (2) + (3) \Rightarrow [SES] = \frac{k_3[S]}{(k_{-3} + \beta k_2)} \cdot \frac{k_1[S][E]}{(k_{-1} + \cancel{k_3[S]} + k_2 - \cancel{k_3[S]})} \quad (4)$$

$$[E] = [E] + [ES] + [SES]$$

$$= [E] + \frac{k_1[ES][E]}{k_{-1} + k_2} + \frac{k_1 k_3 [E][S]^2}{(k_{-3} + \beta k_2)(k_{-1} + k_2)}$$

$$= [E] + \frac{[S][E]}{K_m} + \frac{[E][S]^2}{K_m' \cdot K_m}$$

$$= [E] \left[1 + \frac{[S]}{K_m} + \frac{[S]^2}{K_m' \cdot K_m} \right]$$

$$3) \frac{dP}{dt} = k_2[ES] + \beta k_2[SES] = \frac{k_2[ES][E]}{K_m} + \frac{\beta k_2[E][S]^2}{K_m' \cdot K_m}$$

$$= \frac{k_2 \cdot K_m' \cdot [S][E] + \beta k_2 [E][S]^2}{K_m \cdot K_m'}$$

$$\frac{k_2[S][E]}{K_m} \left(1 + \frac{\beta[S]}{K_m'} \right)$$